

Getting Started with Docker Swarm

In this article, we present Docker Swarm, a native clustering tool that turns a pool of Docker hosts into a single virtual Docker host.

Docker Swarm is just one year and four months old and it has already been accepted as production-ready software. Docker Inc. announced Docker Swarm during DockerConEU in December 2014. Docker Swarm is a clustering manager for Docker, and Docker Machine is a CLI tool for provisioning Docker hosts. Docker Inc. intended to provide a complete and integrated solution for running containers, not allowing itself to be restricted to only providing the Docker engine.

Docker Swarm is the native clustering tool for Docker. It uses the standard Docker API, i.e., containers which can be launched using normal Docker run commands. Swarm takes care of selecting an appropriate host to run the container on. What this actually means is that the other tools that use the Docker API, such as Compose and bespoke scripts, can use Swarm without any changes and take advantage of running on a cluster rather than a single host.

The basic architecture of Swarm is fairly straightforward:

1. Each host runs a Swarm agent and one host runs a Swarm manager (on small test clusters, this host may also run an agent).
2. The manager is responsible for the orchestration and scheduling of containers on the hosts.
3. Swarm can be run in a high-availability mode where etcd, Consul, or ZooKeeper are used to handle failover to a back-up manager.

There are several different methods for how hosts are found and added to a cluster, which is known as ‘discovery’ in Swarm. By default, token-based discovery is used to keep the addresses of hosts in a list stored in the Docker Hub.

Let’s start with the practical implementation of Docker Swarm. I quickly turned on a 4-node cluster on my Google Cloud Engine, one Swarm master node, two agent nodes and one Swarm manager node to manage the overall cluster. The details of my environment are given in Table 1.

Setting up a Swarm master node

Ensure that Docker 1.9.1 is installed on all the nodes of the Docker Swarm cluster. You can use the following command to update all the Docker hosts to the latest version:

```
wget -qO- https://get.docker.com/ | sh
```



Note: It is upper case O and not zero.

The output will look something like what’s shown below:

```
Processing triggers for systemd (225-1ubuntu9) ...
Processing triggers for man-db (2.7.4-1) ...
Setting up docker-engine (1.9.1-0-wily) ...
Installing new version of config file /etc/bash_completion.d/
```

